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How Much Data Do Transformers Need? A Scaling Analysis of DistilBERT for Fake News Detection

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"How Much Data Do Transformers Need? A Scaling Analysis of DistilBERT for Fake News Detection"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Gurpreet Singh**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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Abstract

Fake news has become a significant challenge in the digital era due to its rapid spread through social media platforms, influencing public opinion, health decisions, and political processes. This study explores the effectiveness of different Natural Language Processing (NLP) approaches for fake news detection, including classical machine learning models (Logistic Regression, Support Vector Machine, and Random Forest), a deep learning model (LSTM), and a transformer-based model (DistilBERT).

The research is conducted on the WELFake dataset, containing over 71,000 labeled news articles, using a consistent 80/20 train-test split. A key focus of this work is to analyze the performance of DistilBERT under data and computational constraints, particularly in CPU-only environments. To achieve this, DistilBERT is trained on progressively increasing dataset sizes of 1000, 3000, and 5000 samples, while other models are trained on the full dataset for comparison.

The results demonstrate that all models achieve high performance, with LSTM obtaining the highest accuracy (96.55%) and SVM performing closely (96.46%). Notably, DistilBERT achieves competitive performance even with limited data, reaching 96.00% accuracy with only 3000 samples and showing minimal improvement beyond this point. This indicates that transformer models can effectively leverage pre-trained knowledge to perform well in low-resource settings.

The study highlights that significant performance gains occur with relatively small increases in training data, after which improvements plateau. Additionally, it emphasizes the critical role of preprocessing techniques in enhancing model performance, particularly for classical approaches. Overall, the findings suggest that efficient and practical fake news detection systems can be developed even under limited data and computational resources.

1. Introduction

Fake news has emerged as a critical issue in today's digital world, significantly impacting public opinion, political processes, financial markets, and public health. With the rapid growth of social media platforms, misinformation spreads faster than ever before, often reaching a larger audience than verified information. Events such as the COVID-19 pandemic have clearly demonstrated how false information regarding treatments and vaccines can influence human behavior and decision-making. This growing challenge highlights the urgent need for reliable and scalable fake news detection systems.

Traditional fact-checking methods, although effective, are not sufficient to handle the massive volume of content generated daily. As a result, researchers have increasingly turned to automated approaches using Natural Language Processing (NLP) to identify and classify fake news. Over time, NLP techniques have evolved through three major stages. The first stage involves classical machine learning models such as Logistic Regression, Support Vector Machine (SVM), and Random Forest, which rely on feature extraction methods like TF-IDF. These models are fast and interpretable but lack the ability to capture contextual and sequential relationships in text.

The second stage includes deep learning models such as Long Short-Term Memory (LSTM) networks, which process text sequentially and can capture dependencies across words in a sentence. While LSTMs improve upon classical methods by considering word order, they still face limitations in modeling long-range dependencies and require significant computational resources.

The third and most advanced stage is represented by transformer-based models such as BERT and DistilBERT. These models use attention mechanisms to understand relationships between all words in a sentence simultaneously, enabling them to capture contextual meaning more effectively. DistilBERT, a lightweight version of BERT, is particularly useful in environments with limited computational resources as it retains most of the performance while reducing model size and training cost.

Most existing studies on transformer models assume access to large datasets and high-end hardware such as GPUs. However, such resources are not always available in real-world

academic or low-resource environments. This research addresses this gap by analyzing how the performance of DistilBERT scales with different dataset sizes under CPU-only conditions. Specifically, the model is trained on subsets of 1000, 3000, and 5000 samples to understand how much data is actually required to achieve competitive performance.

In addition to evaluating DistilBERT, this study also compares it with classical machine learning models and LSTM under identical conditions using the WELFake dataset. The objective is not only to compare performance but also to understand the trade-offs between model complexity, data requirements, and computational constraints. This approach provides practical insights into selecting appropriate models for fake news detection, especially in scenarios where resources are limited.

Overall, this study contributes to the field by demonstrating that transformer models can perform effectively even with limited data and computational power, making them more accessible and practical for real-world applications.

2. Methodology

The methodology of this study follows a structured and consistent Natural Language Processing (NLP) pipeline to ensure fair comparison across all models. The overall process includes dataset selection, data preprocessing, feature extraction, model training, and evaluation. All models were trained and tested under identical conditions, including the same dataset split and random seed, to maintain experimental consistency.

2.1 Dataset:

The experiments were conducted using the WELFake dataset, a widely used benchmark for fake news detection. The dataset originally contains 72,134 news articles collected from multiple sources. After removing duplicate and null entries, a total of 71,537 labeled articles were retained. Each article is classified as either real (0) or fake (1). The dataset has a nearly balanced class distribution, which makes accuracy and F1-score reliable evaluation metrics. The dataset was split into 80% for training (57,229 samples) and 20% for testing (14,308 samples).

2.2 Data Preprocessing:

To prepare the data for modeling, several preprocessing steps were applied. The title and body

text of each article were combined into a single content field. The text was then converted to lowercase, and unwanted elements such as URLs, non-alphabetic characters, and extra whitespace were removed. For classical machine learning models, English stopwords were also removed to reduce noise and improve feature quality. In contrast, DistilBERT used its own tokenizer, which handles subword tokenization and requires minimal preprocessing.

2.3 Feature Extraction:

For classical models, text data was transformed into numerical form using the TF-IDF (Term Frequency–Inverse Document Frequency) technique. The configuration included a maximum vocabulary size of 10,000 features, n-gram range of (1,2) to capture both single words and word pairs, and a maximum document frequency of 0.7 to remove overly common terms.

For the LSTM model, an embedding layer was used to learn word representations during training. DistilBERT utilized its pre-trained tokenizer, which converts text into contextualized token embeddings.

2.4 Model Training:

Five models were implemented and trained:

1. **Logistic Regression** – Configured with a maximum of 1000 iterations to ensure convergence.
2. **Support Vector Machine (SVM)** – Implemented using LinearSVC for efficient linear classification.
3. **Random Forest** – Configured with 50 decision trees and parallel processing enabled.
4. **LSTM (Deep Learning Model)** – Consisted of an embedding layer, an LSTM layer with 64 units, a dropout layer (rate 0.5), and a dense output layer with sigmoid activation. The model was trained for 5 epochs with a batch size of 32.
5. **DistilBERT (Transformer Model)** – Fine-tuned using the Hugging Face Transformers library under CPU-only constraints. The model was trained with a maximum sequence length of 64 tokens, batch size of 2, and 2 epochs.

A key aspect of this study is the scaling analysis of DistilBERT. The model was trained on three different dataset sizes—1000, 3000, and 5000 samples—while keeping all other parameters constant. This allowed for a controlled study of how performance changes with increasing data.

2.5 Evaluation Metrics:

All models were evaluated on the same test set using four standard classification metrics: Accuracy, Precision, Recall, and F1-score. Among these, F1-score is particularly important as it balances precision and recall, especially in datasets with near-equal class distribution.

This systematic methodology ensures that the comparison between models is fair and that the observed differences in performance are due to model capabilities rather than variations in experimental setup.

3. Tools and Technologies

The implementation of this project was carried out using a combination of programming tools, libraries, and platforms suitable for Natural Language Processing (NLP) and machine learning tasks.

3.1 Programming Language:

Python was used as the primary programming language due to its simplicity, flexibility, and strong support for data science and machine learning applications.

3.2 Development Platform:

Google Colab was used for executing the code. It provides a cloud-based environment with pre-installed libraries, making it convenient for running experiments without requiring local system setup. The experiments were conducted under CPU-only constraints.

3.3 Libraries and Frameworks:

- **NumPy:** Used for numerical computations and handling arrays efficiently.
- **Pandas:** Used for data manipulation, cleaning, and preprocessing of the dataset.
- **Scikit-learn:** Used for implementing classical machine learning models such as Logistic Regression, SVM, and Random Forest, as well as TF-IDF feature extraction.
- **TensorFlow/Keras:** Used to build and train the LSTM deep learning model.
- **Hugging Face Transformers:** Used to implement and fine-tune the DistilBERT transformer model for text classification tasks.

3.4 Hardware Configuration:

All experiments were performed in a CPU-based environment with approximately 12 GB RAM and no GPU support, which helped in analyzing model performance under limited computational resources.

These tools and technologies together enabled efficient data processing, model training, and evaluation for the fake news detection system.

4. Implementation

The implementation of this project was carried out through a structured pipeline to ensure consistency, reproducibility, and fair comparison across all models. The workflow included data loading, preprocessing, feature extraction, model training, and evaluation.

4.1 Data Loading and Preparation:

The WELFake dataset was loaded using Pandas. Initial inspection was performed to understand the dataset structure and identify missing or duplicate values. All null entries and duplicate rows were removed to maintain data quality. The final dataset consisted of 71,537 labeled news articles. The title and body text of each article were combined into a single content field to capture complete contextual information.

4.2 Data Preprocessing:

Text preprocessing was applied to clean and standardize the data. This involved converting text to lowercase, removing URLs, eliminating non-alphabetic characters, and removing extra whitespace. For classical machine learning models, stopwords were removed to reduce noise and improve feature quality. For the DistilBERT model, minimal preprocessing was performed since its tokenizer handles text processing internally.

4.3 Train-Test Split:

The dataset was divided into training and testing sets in an 80:20 ratio, resulting in 57,229 training samples and 14,308 testing samples. A fixed random state (42) was used to ensure reproducibility across all experiments.

4.4 Feature Extraction:

For classical models, TF-IDF vectorization was used to convert text into numerical features. The configuration included a vocabulary size of 10,000, n-gram range of (1,2), and a maximum document frequency of 0.7.

For the LSTM model, the text was tokenized and padded to a fixed sequence length of 500 tokens, followed by an embedding layer to learn word representations.

For DistilBERT, tokenization was handled using a pre-trained tokenizer with a maximum sequence length of 64 tokens, ensuring efficient computation under CPU constraints.

4.5 Model Implementation and Training:

- **Logistic Regression:** Implemented using Scikit-learn with a maximum of 1000 iterations.
- **Support Vector Machine (SVM):** Implemented using LinearSVC to perform efficient linear classification.
- **Random Forest:** Configured with 50 decision trees, where predictions are made using majority voting.
- **LSTM Model:** Built using TensorFlow/Keras with an embedding layer, an LSTM layer with 64 units, a dropout layer (0.5), and a dense output layer with sigmoid activation. The model was trained for 5 epochs with a batch size of 32 and a validation split of 10%.
- **DistilBERT Model:** Fine-tuned using the Hugging Face Transformers library under CPU-only constraints. The model was trained with a batch size of 2, maximum sequence length of 64 tokens, and 2 epochs.

A key part of the implementation was the scaling analysis of DistilBERT. The model was trained on three different dataset sizes—1000, 3000, and 5000 samples—while keeping all other parameters constant. This allowed a clear understanding of how performance changes with increasing data under limited resources.

4.6 Model Evaluation:

All models were evaluated using accuracy, precision, recall, and F1-score on the same test dataset. This ensured a fair and consistent comparison of model performance.

Overall, the implementation was designed to simulate real-world constraints by using a CPU-only environment while maintaining a rigorous and controlled experimental setup.

5. Outcomes

The results of this study provide a comprehensive comparison of classical machine learning models, deep learning models, and transformer-based models for fake news detection under consistent experimental conditions.

5.1 Overall Model Performance:

All models achieved high performance, with accuracy values lying within a narrow range of approximately 1%. Among all models, the LSTM model achieved the highest accuracy of **96.55%**, followed closely by Support Vector Machine (SVM) with **96.46%**, Random Forest with **96.23%**, and Logistic Regression with **95.58%**. The DistilBERT model, when trained on 3000 samples, achieved an accuracy of **96.00%**, which is highly competitive despite being trained on significantly less data.

This indicates that the WELFake dataset is well-structured and balanced, allowing even simpler models to perform effectively when appropriate preprocessing and feature engineering techniques are applied.

5.2 Comparative Analysis of Model Types:

Classical machine learning models demonstrated strong performance due to effective TF-IDF feature representation. SVM, in particular, showed excellent results, highlighting its capability in handling high-dimensional text data. Logistic Regression, although slightly lower in performance, still achieved competitive results.

The LSTM model outperformed all other models, benefiting from its ability to capture sequential dependencies in text. However, the performance improvement over classical models was marginal, suggesting that for this dataset, complex architectures do not provide a significant advantage.

The DistilBERT transformer model demonstrated strong performance even under constrained conditions. Despite reduced training data and limited computational resources, it performed at a level comparable to models trained on the full dataset. This highlights the strength of pre-trained transformer models in capturing linguistic patterns effectively.

5.3 DistilBERT Data Scaling Results:

A key focus of this study was to analyze how the performance of DistilBERT changes with increasing training data. The results are as follows:

- **1000 samples:** 94.41% accuracy
- **3000 samples:** 96.00% accuracy
- **5000 samples:** 96.07% accuracy

A significant improvement is observed when increasing the dataset size from 1000 to 3000 samples. However, beyond 3000 samples, the improvement becomes marginal. This indicates that most of the learning occurs in the early stages, and additional data contributes only minor gains.

Notably, the performance achieved with 3000 samples (which is approximately 5.2% of the full training dataset) is very close to the performance of SVM trained on the complete dataset. This demonstrates the **data efficiency of DistilBERT**, making it highly suitable for low-resource environments.

5.4 Error Analysis:

The models exhibited some classification errors in distinguishing between real and fake news articles, particularly in cases where both types of articles shared similar vocabulary and writing styles. Classical models showed slightly higher false positives and false negatives compared to LSTM, indicating limitations in capturing contextual nuances.

The LSTM model reduced these errors by better understanding sequential relationships in text. DistilBERT further improved contextual understanding due to its attention mechanism, although its performance was influenced by limited training data and smaller batch size.

5.5 Training Behavior Observations:

The LSTM model showed stable learning behavior, with training accuracy improving steadily over epochs and validation accuracy stabilizing after initial fluctuations. The use of dropout helped in reducing overfitting.

The DistilBERT model showed a consistent decrease in training loss, indicating effective learning even with a small batch size and limited epochs. However, slight fluctuations in loss were observed due to the constrained training setup.

5.6 Impact of Preprocessing:

An important finding of this study is the significant role of preprocessing techniques in improving model performance. Adjustments such as removing stopwords, setting appropriate n-gram ranges, and limiting vocabulary size had a noticeable impact on classical models. In some cases, preprocessing choices influenced performance more than the choice of algorithm itself.

5.7 Practical Outcomes:

The study demonstrates that:

- High accuracy in fake news detection can be achieved using both classical and advanced models.
- Transformer models like DistilBERT can perform efficiently even with limited data and computational resources.
- A dataset size of around **3000 samples** is sufficient to achieve near-optimal performance with DistilBERT.
- Classical models remain strong and practical choices in resource-constrained environments.

Overall, the results provide valuable insights into selecting appropriate models based on data availability and computational constraints, making this study highly relevant for real-world applications and academic research.

6. Conclusion and Future Scope

This study presented a comparative analysis of three generations of NLP models—classical machine learning models, deep learning (LSTM), and transformer-based models (DistilBERT)—for fake news detection using the WELFake dataset. The results showed that all models performed with high accuracy, with LSTM achieving the highest performance, closely followed by SVM and other classical models.

A key contribution of this work is the data scaling analysis of DistilBERT under CPU-only constraints. The findings reveal that DistilBERT achieves competitive performance even with limited training data, reaching near state-of-the-art accuracy with only 3000 samples. Beyond this

point, performance improvements become minimal, indicating that transformer models can effectively leverage pre-trained knowledge and do not necessarily require large datasets to perform well.

The study also highlights the importance of preprocessing techniques, particularly for classical models, where feature engineering significantly impacts performance. Overall, the results demonstrate that efficient and reliable fake news detection systems can be developed even in low-resource environments without access to high-end computational infrastructure.

Future Scope

There are several directions in which this research can be extended. Future work can explore the use of larger and more complex transformer models such as BERT, RoBERTa, or ALBERT to evaluate performance improvements under similar constraints. Experiments can also be conducted using GPU-based environments to compare efficiency and training time.

Another potential extension is to apply the models to more diverse and domain-specific datasets, such as health-related or political misinformation, to evaluate generalization capabilities. Incorporating multilingual datasets can further enhance the applicability of the system across different regions and languages.

Additionally, future research can focus on integrating multimodal data, such as images, videos, and social network information, to improve detection accuracy. Real-time fake news detection systems and deployment in practical applications, such as social media monitoring tools, can also be explored.

Finally, further optimization of preprocessing techniques and hyperparameters can be performed to enhance model efficiency, especially in resource-constrained environments.

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8. Appendices

Appendices

Appendix A: Dataset Description

The dataset used in this study is the WELFake dataset, which is a widely recognized benchmark dataset for fake news detection tasks. It was created by combining multiple reliable sources such as Kaggle, McIntire, Reuters, and BuzzFeed Political to ensure diversity, reduce bias, and improve generalization of the models. The dataset initially contained 72,134 news articles. After performing data cleaning steps such as removing null values and duplicate entries, a total of 71,537 articles were retained for analysis. Each article is labeled as either real (0) or fake (1), and the dataset maintains a nearly balanced class distribution. This balance is important as it prevents

model bias toward a particular class and makes evaluation metrics such as accuracy and F1-score more reliable and meaningful.

Appendix B: Data Preprocessing Steps

The following preprocessing steps were applied systematically to improve data quality and ensure consistency across all models:

- Conversion of all text to lowercase to maintain uniformity
- Removal of URLs to eliminate irrelevant external links
- Removal of non-alphabetic characters such as punctuation and numbers
- Removal of extra whitespace to standardize formatting
- Removal of stopwords for classical machine learning models to reduce noise and improve feature importance
- Tokenization using the DistilBERT tokenizer for the transformer model, which handles subword segmentation and contextual representation

These preprocessing steps played a crucial role in enhancing the effectiveness of feature extraction and improving overall model performance.

Appendix C: Model Configurations

1. Logistic Regression:

- Max iterations: 1000
- Used for binary classification with TF-IDF features

2. Support Vector Machine (SVM):

- Model: LinearSVC
- Efficient for high-dimensional text data and large feature spaces

3. Random Forest:

- Number of estimators: 50
- Random state: 42
- Uses ensemble learning by combining multiple decision trees

4. LSTM Model:

- Vocabulary size: 10,000
- Embedding dimension: 128
- LSTM units: 64
- Dropout rate: 0.5 (to prevent overfitting)
- Batch size: 32
- Epochs: 5
- Validation split: 10%
- Sequence length: 500 tokens

5. DistilBERT Model:

- Model: distilbert-base-uncased
- Max sequence length: 64
- Batch size: 2
- Epochs: 2
- Training samples: 1000, 3000, 5000 (used for scaling analysis)
- Fine-tuned using Hugging Face Transformers library under CPU constraints

Appendix D: Evaluation Metrics

The models were evaluated using standard classification metrics to ensure a comprehensive performance comparison:

- **Accuracy:** Measures overall correctness of predictions
- **Precision:** Measures how many predicted positives are actually correct
- **Recall:** Measures how many actual positives are correctly identified
- **F1-score:** Harmonic mean of precision and recall, especially useful for balanced datasets

These metrics were computed on the same test dataset to ensure fairness and consistency in evaluation.

Appendix E: Hardware and Environment

- Platform: Google Colab

- Execution Mode: CPU-only (no GPU acceleration)
- RAM: Approximately 12 GB
- Programming Language: Python
- Libraries: NumPy, Pandas, Scikit-learn, TensorFlow/Keras, Hugging Face Transformers

This setup represents a realistic and constrained computational environment often encountered in academic and research settings, where access to high-performance hardware such as GPUs is limited. By conducting experiments under such conditions, the study highlights the feasibility of implementing effective fake news detection systems without relying on expensive infrastructure. It further emphasizes that with optimized preprocessing, efficient model selection, and proper use of pre-trained architectures like DistilBERT, strong performance can still be achieved, making the approach practical, scalable, and applicable to real-world scenarios with limited computational resources.